

6	a	i	<u>Number of protons in the nucleus</u>	1
		ii	A high-speed electron	1
		iii	An <u>neutron in the nucleus decays to form a proton and an electron</u> with the latter <u>being ejected out of the nucleus.</u>	1
		iv	The <u>existence of the neutrino</u> to conserve energy and momentum in the decay.	1
	b	i	Radioactive decay is a spontaneous process. It <u>is not affected by external temperature or pressure.</u>	1
		ii	Radioactive decay is a <u>random</u> process.	1
			Half life is the <u>average time taken for half the amount of nuclei to decay</u> , hence it <u>is likely that there will be 0.5 mol left but not definitely.</u>	1
	c		If the experiment is repeated a lot of time, the average amount left will tend to 0.5 mol.	
		i	As the particles travel, they <u>lose energy and slow down.</u>	1
			Since the <u>radius of the circular path is proportional to the velocity of the particles</u> , the radius of the path gets smaller as they travel.	1
		ii	Gamma radiations are <u>weakly ionizing</u> hence they do not produce much bubbles	1